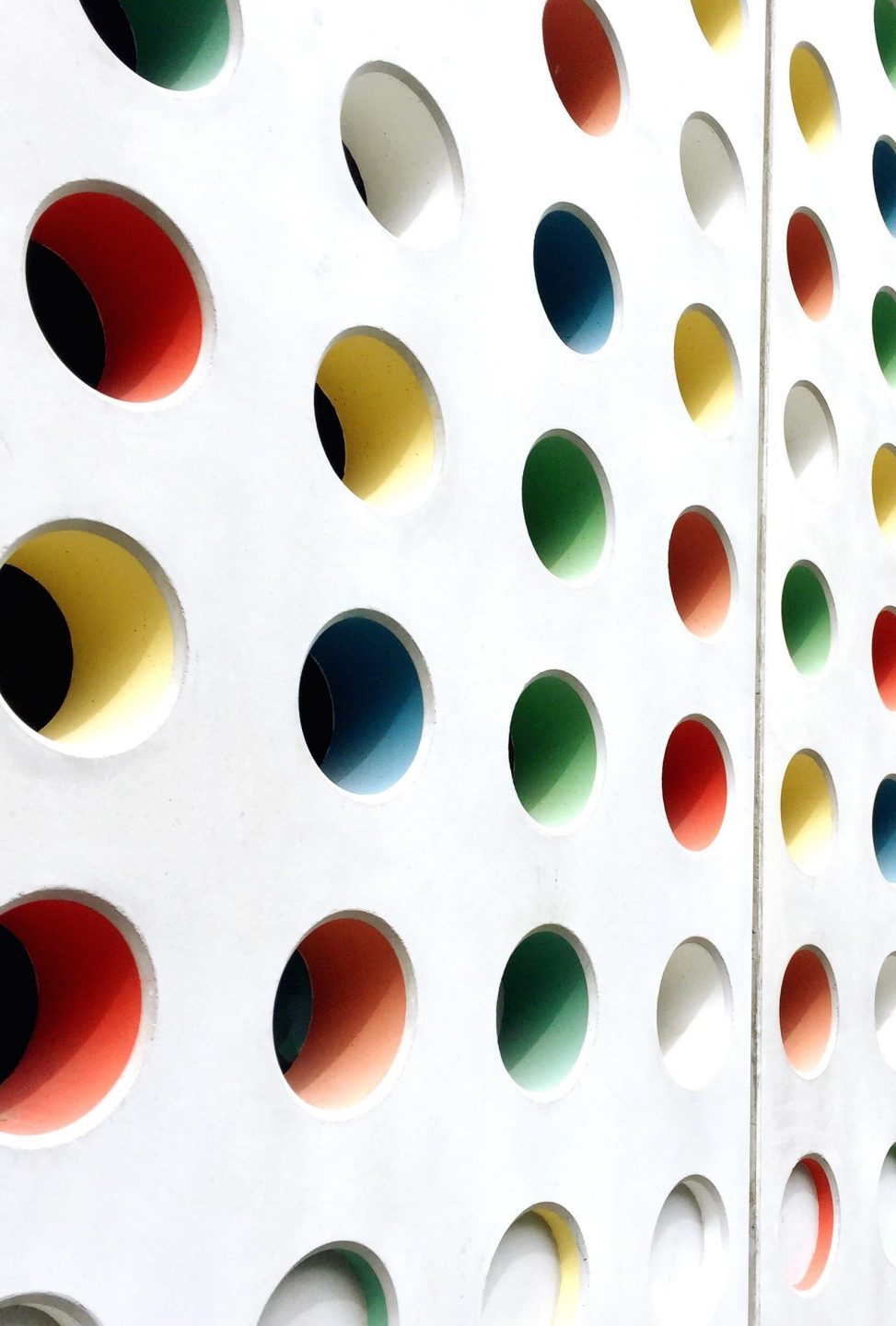




# RDF

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BY

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# DCMES for RDF/XML

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The Dublin Core Metadata Element Set (DCMES) can be represented in many syntax formats. Encode the DCMES in RDF/XML, provides a DTD to validate the documents and describes a method to link them from web pages.

Encoding for the DCMES in XML[XML-SPEC] using simple RDF[RDFMS], provides a DTD and W3C XML Schemas[XMLSCHEMA] to validate the documents.

# Goals of Dublin Core

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1. The primary goal is to provide a simple encoding, where there are no extra elements, qualifiers, optional or varying parts allowed. This allows the resulting data to be validated against a DTD and guaranteed usable by XML parsers.
2. A secondary goal was to make the encoding also be valid RDF[RDFMS] which allows the document to be manipulated using the RDF model. Limit the RDF *constructs* to the minimum, and the result is a standard header and footer for every document.

# Encoding of Dublin Core™ in XML

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Step by step method of how to create a document for the DCMES in XML.

# 1. XML declaration

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Any well-formed XML document must include a statement of the version of XML used (and content encoding). At present, the only valid version of XML, as defined in the W3C Recommendation, is 1.0. Thus the document must start with

```
<?xml version="1.0"?>
```

on the first line.

## 2. Referencing the XML DTD

---

```
<!DOCTYPE rdf:RDF PUBLIC "-//DUBLIN CORE//DCMES DTD 2002/07/31//EN"
```

```
"http://dublincore.org/specifications/dublin-core/dcmes-xml/2002-07-31/dcmes-xml-dtd.dtd">
```

### 3. Declaring the use of RDF

---

It is necessary to declare that RDF[RDFMS] is being used so that applications can recognize this is an RDF/XML document. This declares the outer **rdf:RDF** containing tag with its XML namespace and the XML namespace for the DCMES elements.

```
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/">
```

as the next line in the document, following the XML DTD reference.

## 4. Describing the resources

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Encoding can describe multiple resources in a single document. Each resource described is enclosed in a container element, a pair of `rdf:Description` tags. Resources may have no, one or several identifiers and some of these may be URIs.

If a resource has at least one URI, the most appropriate one should be used as the value of the `rdf:about` attribute of the `rdf:Description` tag like this:

```
<rdf:Description rdf:about="http://example.org/">
```

```
...
```

```
</rdf:Description>
```



# Describing the resources Conti...

---

1. `rdf:Description` container, put each of the Dublin Core elements with the `dc:namespace` prefix before it, so for example the `Title` element becomes `dc:title` all in lowercase and used inside the `rdf:Description` container like this given below:

```
<rdf:Description rdf:about="http://example.org/">
  <dc:title>My Home Page</dc:title>
</rdf:Description>
```

2. If the value of the Dublin Core element is a resource which has a URI rather than plain text, it should be recorded in the value of the `rdf:resource` attribute on the tag, with empty tag content. For example, if the value of the source was a URI, it would be

recorded like this:

```
<rdf:Description rdf:about="http://example.org/">
  <dc:source rdf:resource="http://example.org/elsewhere/" />
</rdf:Description>
```

# Describing the resources Conti...

---

3. There may be more than one *Identifier* element for the resource containing URIs or other identifiers. If a URI identifier is available and appropriate to use, it should be made the value of the `rdf:about` attribute of the `rdf:Description` element. The other *Identifier* element values should be encoded in the same manner as the other elements as described below.

```
<rdf:Description>
  <dc:title>Internet Ethics</dc:title>
  <dc:creator>Duncan Langford</dc:creator>
  <dc:format>Book</dc:format>
  <dc:identifier>ISBN 0333776267</dc:identifier>
</rdf:Description>
```

# s. Language and character encoding

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XML provides an `xml:lang` attribute that can be used on any element. This provides a way to describe the language used for the content of the element. The DCMES provides a Language element which is used to describe the language of the resource.

The values of the elements and attributes will need to be encoded using the rules of XML when there are special characters in the value. The special characters that need to be encoded, and when they need to be are summarized here for reference:

| Plain text                    | XML Encoding | Required in                  |
|-------------------------------|--------------|------------------------------|
| &                             | &amp;        | Element and attribute values |
| <                             | &lt;         | Element and attribute values |
| >                             | &gt;         | Element and attribute values |
| ' (apostrophe / single quote) | &apos;       | Attribute values             |
| " (double quote)              | &quot;       | Attribute values             |

## 6. Finishing off the document

---

The final thing that needs to be done is to close the `rdf:RDF` element opened at the top of the document by adding the following line:

```
</rdf:RDF>
```

```
<?xml version="1.0"?>
<!DOCTYPE rdf:RDF PUBLIC "-//DUBLIN CORE//DCMES DTD 2002 01 24//EN" "http://dublincore.org/specifications/dublin-core/dcmes-xml/2002-01-24/dcmes-xml-dtd.dtd">
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
    xmlns:dc="http://purl.org/dc/elements/1.1/">
  <rdf:Description rdf:about="http://www.ilrt.bristol.ac.uk/people/cmdjb/">
    <dc:title>Dave Beckett's Home Page</dc:title>
    <dc:creator>Dave Beckett</dc:creator>
    <dc:publisher>ILRT, University of Bristol</dc:publisher>
    <dc:date>2000-06-06</dc:date>
  </rdf:Description>
</rdf:RDF>
```

# Example 1:

## Example 2:

---

```
<?xml version="1.0"?>
<!DOCTYPE rdf:RDF SYSTEM "http://dublincore.org/2000/12/01-dcmes-xml-dtd.dtd">
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
        xmlns:dc="http://purl.org/dc/elements/1.1/">
  <rdf:Description rdf:about="http://dublincore.org/">
    <dc:title>Dublin Core™ Metadata Initiative - Home Page</dc:title>
    <dc:description>The Dublin Core™ Metadata Initiative Web site.</dc:description>
    <dc:date>1998-10-10</dc:date>
    <dc:format>text/html</dc:format>
    <dc:language>en</dc:language>
    <dc:contributor>The Dublin Core™ Metadata Initiative</dc:contributor>
    <!-- My guess at the French for the above Dave -->
    <dc:contributor xml:lang="fr">L'Initiative de métadonnées du Dublin Core</dc:contributor>
    <dc:contributor xml:lang="de">der Dublin-Core Metadata-Diskussionen</dc:contributor>
  </rdf:Description>
</rdf:RDF>
```

# W3C XML Schema for Simple Dublin Core™ Metadata Element Set 1.1 in RDF/XML [For Information Only]

There are three XML namespaces involved here and correspondingly there are three W3C XML Schema documents:

[dcmes-xml-xsd.xsd](#) (Main schema)

[dcmes-dc.xsd](#) (Dublin Core™ Element Set)

[dcmes-rdf.xsd](#) (RDF syntax)

# Check and Visualize RDF Documents

<https://www.w3.org/RDF/Validator/>

**Check and Visualize your RDF documents**

[olde servlet](#)

Enter a URI or paste an RDF/XML document into the text field above. A 3-tuple (triple) representation of the corresponding data model as well as an optional graphical visualization of the data model will be displayed.

Check by Direct Input

```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/">
  <rdf:Description rdf:about="http://www.w3.org/">
    <dc:title>World Wide Web Consortium</dc:title>
  </rdf:Description>
</rdf:RDF>
```

Parse RDF

Restore the original example

Clear the textarea

Display Result Options:

Triples and/or Graph: Triples Only ▼

Graph format: PNG - embedded ▼

Paste an RDF/XML document into the following text field to have it checked. More options are available in the [Extended interface](#).



# Result Validation in Triples

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## Validation Results

Your RDF document validated successfully.

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## Triples of the Data Model

| Number | Subject                                             | Predicate                                                                                 | Object                      |
|--------|-----------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------|
| 1      | <a href="http://www.w3.org/">http://www.w3.org/</a> | <a href="http://purl.org/dc/elements/1.1/title">http://purl.org/dc/elements/1.1/title</a> | "World Wide Web Consortium" |

---

## The original RDF/XML document

```
1: <?xml version="1.0"?>
2: <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
3:   xmlns:dc="http://purl.org/dc/elements/1.1/">
4:   <rdf:Description rdf:about="http://www.w3.org/">
5:     <dc:title>World Wide Web Consortium</dc:title>
6:   </rdf:Description>
7: </rdf:RDF>
8:
```

---

# Validate results in Triples and Graphs

## Validation Results

Your RDF document validated successfully.

### Triples of the Data Model

| Number | Subject                                             | Predicate                                                                                 | Object                      |
|--------|-----------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------|
| 1      | <a href="http://www.w3.org/">http://www.w3.org/</a> | <a href="http://purl.org/dc/elements/1.1/title">http://purl.org/dc/elements/1.1/title</a> | "World Wide Web Consortium" |

### The original RDF/XML document

```
1: <?xml version="1.0"?>
2: <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
3:   xmlns:dc="http://purl.org/dc/elements/1.1/">
4:   <rdf:Description rdf:about="http://www.w3.org/">
5:     <dc:title>World Wide Web Consortium</dc:title>
6:   </rdf:Description>
7: </rdf:RDF>
8:
```

### Graph of the data model



# Task to do...

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You have to made RDF file of your assigned task (selected by individual student in XML case). Validate via triple and Graph as well.